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Picosecond time-resolved pure-rotational coherent anti-Stokes Raman spectroscopy in sooting flames

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Abstract

The development of time-resolved dual-broadband pure-rotational coherent anti-Stokes Raman spectroscopy using picosecond laser pulses is investigated for use in the study of highly sooting flames. Axi-symmetric ethylene and propane diffusion flames were studied. Suppression of nonresonant and Raman resonant interference signals is demonstrated by delaying the probe pulse beyond the temporal envelope of the pump pulses, and these spectra are compared with those obtained using conventional polarization-based interference suppression techniques. Flame profiles for both temperature and the O_2 to N_2 ratio are obtained. Evaluated temperatures are corrected for delay-induced spectral heating and compared to published thermocouple measurements.
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Keywords: Time-resolved CARS; Thermometry; Soot; Combustion

1. Introduction

There is currently a need for experimental validation of combustion models as concerns of climate change and harmful pollutant reduction take center-stage in environmental policy. Spectroscopic techniques such as laser-induced fluorescence (LIF) and coherent anti-Stokes Raman spectroscopy (CARS) have been developed for over three decades to probe combustion environments in a nonintrusive manner and gain information about temperature and species concentrations. Spatial and temporal resolution as well as high accuracy and precision are of paramount importance. Recent reports indicate that a reduction in global soot production may be effective in curbing climate change [1]. Therefore, both accurate soot formation models and experimental validation are needed to more fully understand the process by which soot is generated during combustion. Techniques that can be applied to highly sooting flames then must be explored.

CARS is a widely used technique for nonintrusive thermometry and major species concentration determination in flames. In vibrational CARS, an excited vibrational state of a molecule, typically nitrogen in air-fed combustion processes, is excited with a pair of photons, the difference

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frequency of which is in resonance with the vibrational transition. Following the excitation of this coherence, a narrowband probe laser is scattered from the excited molecules and coherent signal is generated. Temperature is determined by fitting the ro-vibrational CARS spectrum. Vibrational CARS has been shown to be especially accurate at temperatures above 1200 K when the first "hot-band" becomes apparent in the signal. This band originates from thermal excitation of a significant fraction of molecules into the first vibrational state, such that the lasers are actually driving the $v = 1 \rightarrow v = 2$ transition, and the ratio of signal from the fundamental and first "hot-band" is quite dependent on temperature at these temperatures.

Another approach is pure-rotational CARS. The distinction here is that the difference frequency between the two excitation beams excites S-branch rotational Raman transitions. Vibrational energy levels are so specific to a molecule that vibrational CARS typically probes only one or two species in the flame, unless multiple pump lasers are used. However, rotational transition frequencies among many molecules important in combustion environments, such as N_2, O_2, CO_2, and CO, fall within a narrow spectral range, allowing for single-shot acquisition of spectra from multiple species [2]. This co-location potentially enables the determination of relative species concentrations. The majority of rotational CARS studies have been at relatively low temperatures [3,4], however, several studies have been performed demonstrating its applicability as a flame–temperature combustion diagnostic [5–7]. There exist some advantages to the RCARS technique over its vibrational CARS counterpart under certain circumstances. First, the wide spacing of the rotational lines in a pure-rotational CARS spectrum mitigates the effect of coherent line-mixing which complicates vibrational CARS at high pressures [2]. Secondly, spectroscopy in highly sooting flames is complicated by Swan band emission from laser-produced C_2, and one of these emission frequencies occurs very near to the typical vibrational CARS signal wavelength of 473 nm when using nitrogen as the probed molecule. This complication can be circumvented by the use of dual-pump vibrational CARS [8,9], in which the probe beam scattering from the vibrational coherences is obtained from a narrowband tunable dye laser, and the frequency location of the signal is adjusted to avoid C_2 interference. This technique, of course, requires a third laser, adding complexity to the standard two-laser CARS setup. Lastly, in principle the RCARS approach allows for multiple species to be monitored in a single shot.

In dual-broadband RCARS, developed in 1986 [10,11], a broadband dye laser serves as the source for both excitation beams. Because photon pairs originate from combinations of frequencies within this bandwidth the effect of laser mode fluctuations is minimized and higher precision and accuracy are achieved [10–12]. Time-resolved measurements are not possible utilizing a conventional nanosecond-based laser system because rotational coherences dephase on a timescale that is significantly shorter than the pulse width. For instance, the exponential decay time constant in RCARS signal due to collisional dephasing was previously reported to be about 60 ps for nitrogen RCARS at room temperature [13].

Although the RCARS approach avoids interference from C_2 in highly sooting flames, there still exists complicating interferences to the signal. As a third-order nonlinear process, CARS signal is proportional to the modulus of the third-order susceptibility tensor squared, $|\chi^{(3)}_{\mathrm{CARS}}|^2$. There exists both resonant and nonresonant contributions to $\chi^{(3)}_{\mathrm{CARS}}$. The electronic nonresonant susceptibility, χ_{nr}, generates signal that interferes coherently with the resonant signal, originating from the resonant susceptibility, χ_r.

$$\chi_r = \sum_J \frac{A_{J,J+2}}{\omega_{J,J+2} - \omega_1 + \omega_2 - ip\gamma_{J,J+2}}. \tag{1}$$

In Eq. (1) J is the rotational quantum number of the lower level, $A_{J,J+2}$ is the amplitude for the $J \rightarrow J+2$ transition, ω_1 and ω_2 are the frequency components of the pump laser exciting the rotational coherence, $\omega_{J,J+2}$ is the transition frequency between the rotational energy levels $J \rightarrow J+2$, p is the pressure and $\gamma_{J,J+2}$ is the Raman linewidth of the transition (half width at half maximum). The gas phase temperature affects the relative intensities of the J-lines in an RCARS spectrum, as the relative population of rotational levels will follow a Boltzmann distribution in temperature and affect the value of $A_{J,J+2}$ in Eq. (1).

Another common source of interfering signal in RCARS of fuel-rich environments is smeared vibrational CARS. SV-CARS originates when the difference frequency between the probe laser ω_3 and the pump laser coincides with the energy of a vibrational mode in Raman-active molecules. The difference frequency excites a vibrational coherence from which the second broadband pulse scatters, generating a vibrational CARS signal that is "smeared" out by the broad spectral width of the pulse. This SV-CARS signal can be eliminated by a careful selection of laser dye such that the difference frequency between the probe and the dye laser frequency is placed where no vibrational modes would be expected. However, even with this goal in mind, a recent work [14] demonstrated that contributions from SV-CARS could not be completely eliminated in a flame, and a polynomial was fit to the baseline and subtracted to achieve good theoretical fits to the data.

The major approach in RCARS for the removal of nonresonant background from the signal is based on polarization selection rules [14]. It can be shown that the general equations for RCARS relating the polarization of the three incoming beams to the polarization of the signal are given by Vestin et al. [14]:

$$\tan(\beta) = -\frac{2\cos(\theta)\sin(\phi) - 3\sin(\theta)\cos(\phi)}{4\cos(\theta)\cos(\phi) + 3\sin(\theta)\sin(\phi)}, \quad (2)$$

$$\tan(\delta) = \frac{\cos(\theta)\sin(\phi) + \sin(\theta)\cos(\phi)}{3\cos(\theta)\cos(\phi) + \sin(\theta)\sin(\phi)}, \quad (3)$$

where β is the polarization angle at which the resonant signal is emitted, δ is the polarization angle for the nonresonant signal, θ is the angle between the polarizations of the pump beams, and ϕ is the polarization angle between the probe and the pump beam which defines the coordinate origin. Thus, identifying a combination of incoming polarizations yielding β and δ orthogonal to each other allows a polarization analyzer to be placed in the signal path which transmits the highest fraction of resonant signal while fully suppressing the nonresonant signal. Figure 1 plots these equations for parallel pump beams. Polarization-based techniques can also be used to eliminate much of the interfering signal from SV-CARS. Because of the isotropic character of the excited vibrational coherence, using an orthogonally polarized pump or probe beam should suppress much of the SV-CARS signal, leaving only the anisotropic components to contribute.

In this work, we report the use of time-resolved picosecond RCARS spectroscopy as a technique for removing unwanted nonresonant and Raman resonant signals in the highly sooting jet diffusion flame of a Santoro-type burner frequently used in sooting flame studies. RCARS spectra are compared to the polarization-based technique for both interference suppression and signal reduction. In time-resolved RCARS, the two excitation pulses arrive at the same time to drive the rotational Raman coherences, but the probe pulse is delayed in time beyond the envelope of the pump pulses. The electronic nonresonant signal has such short lifetime that its signal decays with the pulse width of the pump pulses [13,15] when picosecond pulses are used. N_2 and O_2 rotational coherences, however, have lifetimes which are long compared to the pulse width of a picosecond laser allowing for delayed probing. Delayed-probe picosecond RCARS has been demonstrated to significantly suppress nonresonant signal in as little as a 150 ps probe delay [13]. A significant benefit of delayed-probe RCARS is that the ordering of the pulses is defined by the experiment. Because the 532 nm probe pulse arrives after the broadband dye-laser pulses, no possibility exists for the probe pulse to mix with a pump pulse and drive a vibrational coherence. Thus the time-ordering of the pulses eliminates any contribution from SV-CARS once the probe has been delayed beyond the temporal envelope of the two pump pulses [16]. Complete suppression of both the nonresonant background and SV-CARS signal is simultaneously demonstrated in a sooting flame and compared to polarization-based techniques.

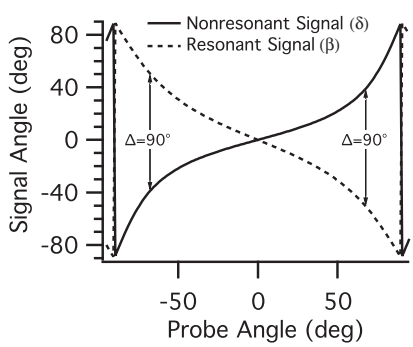

Fig. 1. Polarization angle of the emitted nonresonant (solid) and resonant (dashed) signal for parallel pump beams as function of probe beam polarization angle. Vertical arrows indicate probe polarization angles producing orthogonal β and δ angles.

2. Experimental

The flame measurements were performed using a sooting laminar diffusion flame on a Santoro burner. Commonly used for sooting flame studies [17,18], the burner consists of a brass fuel tube with a 1.1-cm inner diameter surrounded by ceramic honeycomb with a 10.2-cm outer diameter for a co-flow of air. The flame is stabilized with a chimney 19.5 cm tall and 10.2 cm in diameter. A well-characterized ethylene diffusion flame [17] was used with flow rates of 0.23 standard liters per minute (SLM) ethylene and 42 SLM air for the co-flow. The flow rates were calibrated to within 1% using an MKS Type GBROR flow verifier and mass flow controllers MKS 1479A and MKS 1559A for the low and high flow rates, respectively. The Santoro burner was mounted on a translation stage to move the flame with respect to the RCARS probe volume. Flame profiles were measured by moving the assembly parallel with the beam propagation direction. Although higher spatial resolution could have been achieved by translating the flame perpendicularly to the beams, such movement was difficult to achieve with the current flame setup because the chimney openings to allow transmission of the beams were made as small as possible to ensure the stability of the flame. A second set of profiles using 0.14 SLM propane as the fuel were also obtained.

The second harmonic of a regeneratively amplified Nd:YAG laser operating at 20 Hz was

used both as the source of the probe beam at 532 nm and to pump a broadband dye laser operating at 633 nm using DCM dye. The pulse width of the Nd:YAG laser was approximately 60 ps. The dye laser consisted of an ASE source, followed by two side pumped amplification stages. Typical pulse energies used were 1.0 mJ/pulse for the 532 nm probe beam, and 1.3 mJ/pulse for each of the two 633 nm broadband pump beams. A half-wave plate and polarizing beam splitter was used to separate the red beam into two equal beams for use as the RCARS pump beams. A spherical 300 mm focusing lens two inches in diameter was used to focus all three laser beams to a point with a diameter of approximately 50 μm. The probe volume of approximately 1 mm in length (FWHM) was measured by translating a glass cover slip through the beam crossing region and monitoring the four-wave mixing signal. One of the pump beams was placed on a mechanical translation stage to zero the temporal overlap of the beams, and an electronically controlled mechanical delay stage was placed in the probe beam path to allow precise delay timing of the probe pulse with respect to the two pump pulses ranging from +250 to −1100 ps. A planar BOXCARS phase matching geometry was used such that the signal was emitted nearly collinearly with one of the red pump beams. Following the beam crossing, the green probe beam and the nearly collinear pump beam were directed to a beam dump using a right-angle prism. Three dichroic mirrors were used to separate the signal from the other 633-nm beam. A holographic notch filter was used in some cases in the signal collection path to eliminate stray 532-nm light. A 75-mm focal-length cylindrical focusing lens was used to focus the signal into the spectrometer slit. The slit was controlled with a micrometer and was set to 40 μm. An 1800 groove/mm diffraction grating was used in a 1-m spectrometer (SPEX-1000M) to disperse the signal onto a back-illuminated CCD camera with 24 μm pixel spacing, providing a dispersion of 0.44 cm^{-1} per pixel as determined during the spectral fit of room air.

For the polarization experiments, an approach reported in Ref. [14] was used. The polarization of each of the three beams was initially set to vertical by the use of half-wave plates and verified with a thin-film polarizer. The first 633-nm pump beam was kept vertical and defined the angular origin for the polarization experiments. A half-wave plate was placed in the probe beam path to set ϕ (see Eqs. (2) and (3)), and a half-wave plate was placed in the second pump beam path to set θ. A polarization analyzer was placed in the signal path, and finally a half-wave plate was placed before the spectrometer to rotate the signal to maximize signal passage through the detection system.

Spectra were fitted using a frequency-domain contour-fitting procedure that has been used extensively for nanosecond-laser-based RCARS [19]. The calculation of the CARS spectra and the temperature evaluation were based on a computer code that compares the experimental spectra with a spectrum from a precalculated library by the use of a contour-fitting procedure [20]. The experimental spectra were corrected for the bandwidth profile of the broadband dye laser by dividing the measured spectra by a spectrum of pure argon. Because argon has no rotational transitions, the spectrum is entirely composed of nonresonant signal. The evaluation procedure utilizes the Levenberg–Marquardt algorithm with interpolation between the spectra of the precalculated library by means of cubic splines. The parameters for the experimental slit function are described by a Voigt profile and were determined by fitting experimental spectra taken from air under ambient conditions. These values were used as fixed input parameters for the calculation of the theoretical spectra. For the calculation of the Raman linewidth we used the modified exponential gap law [21]. The parameter sets used can be found in Ref. [22].

3. Results

Figure 2 displays an averaged spectrum taken at the radial center of the propane flame at probe delays of 0 and 150 ps. A delay of only 150 ps is sufficient to completely suppress both the nonresonant background and the SV-CARS contributions to the signal. A signal reduction of 36% is seen in this case. The reduction in signal for this delay is due to collisional dephasing of the rotational coherences and will thus be dependent on the gas temperature. The signals were corrected for dark counts only, and not normalized by the spectrum of argon so as to demonstrate the observed signal counts per shot. For comparison, spectra were taken using polarization techniques to suppress the nonresonant background by placing the probe beam at 67° with respect to the pumps. This creates a 90° polarization difference between the resonant and nonresonant signal, so the polarization analyzer can discriminate against the nonresonant background, while transmitting the resonant signal. Figure 3a displays the spectrum obtained in this manner. As can be seen in the increasing slope on the low energy side of the spectrum, there still exists a significant contribution from the SV-CARS interference. In previous work using the polarization suppression approach [14], the SV-CARS signal was also reported to be present, and the subtraction of a polynomial fit to the baseline was required. It is advantageous then that the time-resolved delayed-probe technique of suppressing the nonresonant signal simultaneously removes the SV-CARS interference. The polarization-based

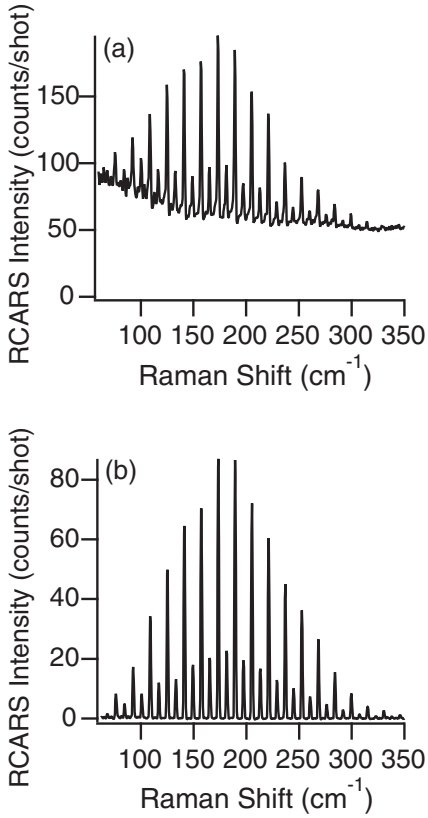

Fig. 2. Averaged spectrum taken at the radial center of a propane diffusion flame at probe delays of (a) 0 ps and (b) 150 ps. The evaluated temperature was 1205 K.

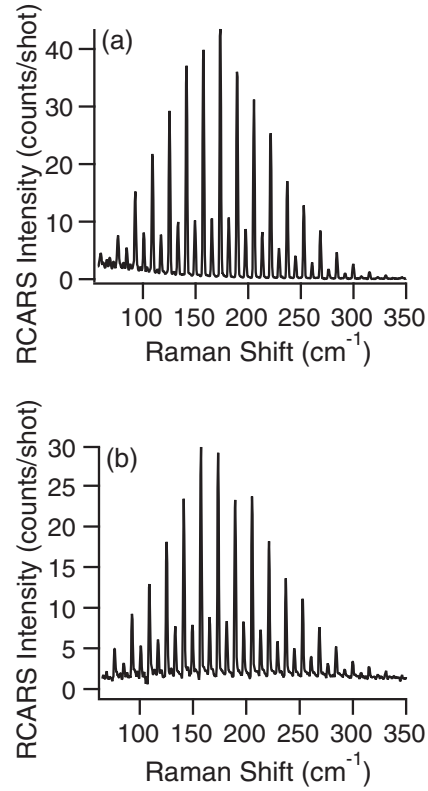

Fig. 3. Averaged spectrum taken at the radial center of a propane diffusion flame using polarization-based techniques to suppress (a) the nonresonant background and (b) the SV-CARS signal. The temperature was evaluated to be 1205 K from Fig. 2b.

approach for the suppression of the nonresonant signal interference was also shown to decrease signal levels by a factor of 61% [14] for a probe beam at 60° with respect to the pumps, while the time-resolved method decreased signal levels by only 36%. Perfect suppression of the nonresonant background occurs with the probe at 67° with respect to the pumps, as seen in Fig. 1, and resulted in a signal attenuation of 68% in this work. Figure 3b shows the spectrum taken in the same location with a probe polarization angle of 90° so as to suppress the isotropic contributions to the SV-CARS signal. This configuration cannot fully remove the SV-CARS signal due to the anisotropic vibrational contribution from the fuel molecules, and nonresonant background is again transmitted with the resonant signal.

RCARS profiles were taken radially in the ethylene jet diffusion flame using a probe delayed by 240 ps, which simultaneously removes contributions from the nonresonant background and SV-CARS. During the time-interval between the excitation of the rotational Raman coherences and the scattering of the probe from these molecules, the molecules undergo dephasing collisions. The effect of these collisions is more significant at the low rotational levels, and decreases with increasing J [23]. As such, the low J-lines in an RCARS spectrum will dephase more quickly than the high J-lines upon a probe delay, making the resulting spectrum appear "hotter" if a time-independent frequency-domain analysis is used that does not explicitly take account of the dephasing rates. A reliable way of getting an error estimate is to compare thermocouple measurements in a temperature controlled oven to the evaluated RCARS temperature. Oven measurements were performed in our laboratory in the temperature range 295–1700 K. The largest deviation between the thermocouple measurements and evaluated RCARS spectra was 55 K for spectra recorded at zero probe delay, giving an indication of the overall accuracy of the technique for high temperatures. Spectra at various probe delays were also taken in the oven and evaluated with the time-independent frequency-domain analysis, allowing for an estimate to be made as to the apparent "heating" of the spectra with probe delay. In Fig. 4 the "corrected" RCARS temperatures take

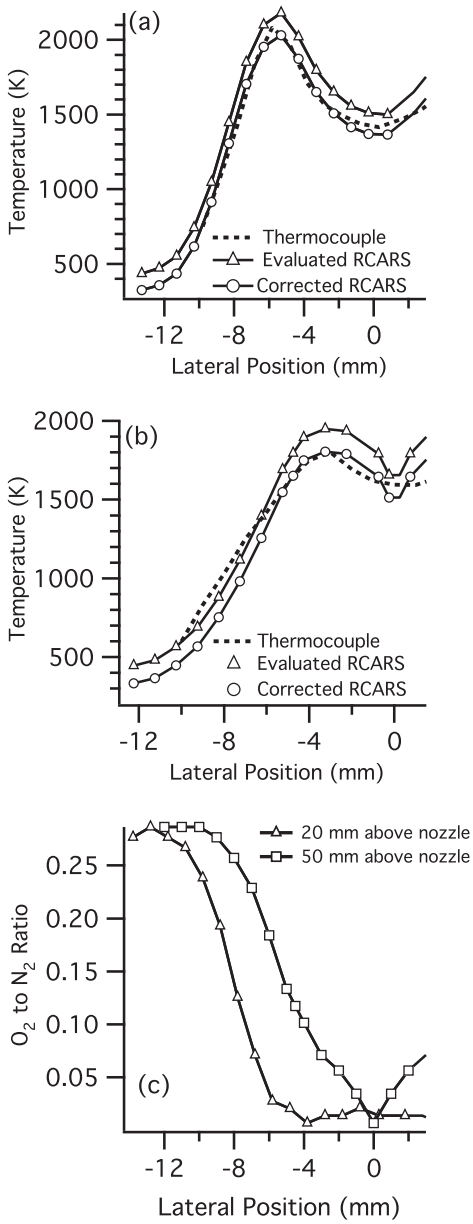

Fig. 4. Temperature profile from a height of (a) 20 mm and (b) 50 mm above the fuel-tube exit of the Santoro burner. (c) Relative O_2 to N_2 ratio profiles.

account of this correction to the evaluated temperatures. It should be noted that these correction factors were calculated based on the dephasing rates of pure N_2 in an oven; however, the flame composition varies from nearly pure ethylene at the flame-center to room air outside the flame-front, with water and CO_2 in the exhaust region. Thus, nitrogen's collision partners will vary from ethylene to O_2 and N_2 depending on the position in the flame. Linewidth measurements for O_2 perturbed by N_2 show J-dependence nearly identical to that of O_2 self-broadened [24]. Similarly, the J-dependence of N_2 linewidth for N_2–CO [25], N_2–CO_2 [26], and N_2–C_2H_2 [27] all follow similar decreasing linewidth with increasing J, and thus will have similar trends in dephasing rates. However, N_2 line broadening by H_2O has been shown [28] to have a much different J-dependence with the intermediate J-values corresponding to the largest linewidth. A time-domain RCARS model is currently under development in our laboratory which explicitly takes into account the J-dependent dephasing rates for time-resolved experiments. Differential dephasing will further affect the calculated relative oxygen to nitrogen ratio as the dephasing rates for nitrogen and oxygen are different [29]. Figure 4c displays the O_2 to N_2 ratio. The ratio was calculated from the spectra at a 240 ps delay. Based on the oven measurements at temperatures from 295 to 1700 K, a 240 ps delay was seen to decrease the O_2 signal roughly 1.35 times more than the N_2 signal due to the faster dephasing rate of O_2. Thus, the calculated ratio profile displayed in Fig. 4c was corrected for this delay-induced ratio change. As can be seen, the ratio at points furthest out from the flame-center reaches the expected ratio for atmospheric air. Again, an RCARS model which explicitly accounts for the species-specific dephasing rates will account for this in the spectral evaluation procedure.

Temperature measurements have previously been reported for the ethylene diffusion flame on the Santoro burner employed in these experiments [17] using the rapid-insertion thermocouple technique. Figure 4 displays the temperature profile from the thermocouple measurements overlaid on the profiles obtained by fitting the RCARS spectra taken at different lateral locations in the flame. The agreement was quite good between the thermocouple measurements in the ethylene flame from the measurements of Santoro and Miller [17] and the delay-corrected temperatures obtained from the time-resolved RCARS measurements for a flame height of 20 mm. The largest deviation was 70 K located at the radial center of the flame in this profile. The stated uncertainty of the thermocouple measurements was estimated to be ± 100 K, and several factors in the thermocouple technique used may contribute to this estimate. One hundred and twenty-five micrometers of diameter thermocouple wires was used, and wires of a diameter this large have been shown [30] to allow significant thermal conduction, causing systematic errors in the measurement. Further, soot-deposition related changes have been shown to affect the accuracy of the rapid-thermocouple-insertion technique [30], estimated to have a 60 K effect at a soot fraction of 6 ppm. The profiles taken at a height of 50 mm held similar agreement

within the flame-front. However, our results begin to differ significantly at positions farther out from the flame. The difference may be due to slight flame instabilities which would be exaggerated at this height in the flame.

A series of temperature profiles were taken in a propane diffusion flame on the Santoro burner using 0.14 SLM propane as the fuel and a 65 SLM co-flow of air and the results are shown in Fig. 5. The dominating interference shown in Fig. 2 for this flame was completely suppressed with a 150-ps delayed probe. Spectra of high quality were obtained for all positions recorded in this flame and evaluated temperatures were corrected for the 150 ps probe delay based on oven measurements. The profiles qualitatively demonstrate the expected rise in flame-center temperature at higher positions in the flame.

4. Discussion

Suppression of nonresonant background was achieved by taking advantage of the relatively long dephasing lifetime of the rotational coherences in flame molecules such as N_2 and O_2 when compared to the lifetime of the nonresonant excitation process. Further, the required time-ordering of the pulses to generate interfering SV-CARS signal necessitates that the green pulse come before or during the 633-nm pump pulses, and delaying past the pulse envelope eliminates this contribution. However, the collisional dephasing which occurs during the time-interval between the pump pulses driving the rotational coherence and the probe pulse interrogating it is highly J-dependent and species dependent. For instance, Fig. 6a illustrates the J-dependence of dephasing rates for pure nitrogen at room temperature and ambient pressure. As can be seen, the exponential decay time constant for molecules with J'' (lower rotational state) of 18 is nearly double that of molecules with $J'' = 4$. As a result, the low J lines in an RCARS spectrum will have relatively lower signal as compared with the high J lines upon a probe delay, which makes probe-delayed spectra appear hotter in a frequency-domain analysis. For the flame measurements reported here, this artificially high temperature was corrected by using the standard frequency-domain analysis on oven measurements with a series of delays. As seen in Fig. 4, a 240 ps probe delay caused an increase in temperature of between 111 and 148 K over the evaluated spectra at a probe delay of zero depending on the temperature. Greater accuracy in the exact value of this correction for flame measurements would necessitate consideration of the change in J-dependent

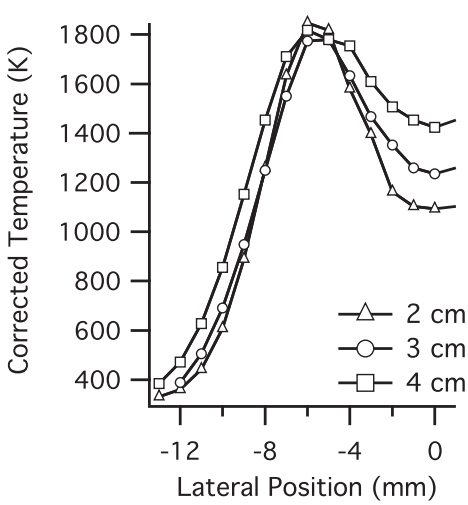

Fig. 5. Temperature profiles calculated from RCARS spectra in a laminar propane diffusion flame with 0.14 SLM propane and 65 SLM air co-flow taken at 2, 3 and 4 cm height.

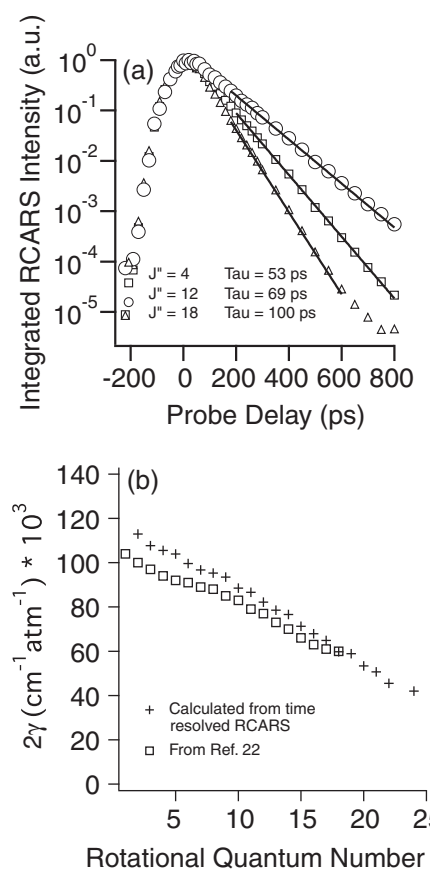

Fig. 6. (a) Integrated RCARS signal as a function of J'' line and probe delay for nitrogen at 295 K. (b) Conversion of J-dependent collisional dephasing times to Raman linewidth for nitrogen at 295 K.

dephasing rate caused by different collision species.

These time-resolved measurements are a direct probe of collisional dephasing, as can be seen in Fig. 6b. Calculation of J''-specific pure rotational Raman linewidths from the exponential decay time-constants determined from each rotational line yields excellent agreement with previously published high resolution N_2 Raman linewidth measurements [22,23]. The linewidths taken from Ref. [22,23] were Q-branch linewidths, and the relationship $\gamma_{J,J+2} = \gamma_{J,J} + \gamma_{J+2,J+2}/2$ was used to calculate the pure-rotational linewidths displayed in Fig. 6b.

5. Summary

Time-resolved dual-broadband RCARS using picosecond pulses was demonstrated as a valuable flame diagnostic for thermometry and major species concentration measurements. The ability to completely suppress simultaneously both interfering contributions from the nonresonant background and SV-CARS was demonstrated in a commonly used laminar nonpremixed sooting diffusion flame with a signal reduction of only 36% in this case. This technique was compared experimentally to the polarization-based approach for suppression of nonresonant background and SV-CARS. It was previously reported that the polarization-based approach for complete suppression of the nonresonant background reduces RCARS signal by 61% [14], and simultaneous suppression of SV-CARS signal is not possible. The delayed probe causes the observed spectra to appear slightly hotter; the magnitude of this effect was estimated by studies in an oven. The corrected temperature profiles were compared to thermocouple measurements in an ethylene diffusion flame and good agreement was found. An RCARS model which explicitly accounts for the collisional dephasing during the time-interval between pump and probe is necessitated and currently under development at Sandia.

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References

[1] J.M. Haywood, V. Ramaswamy, *J. Geophys. Res.* 103 (1998) 6043.
[2] F. Vestin, P.-E. Bengtsson, *Proc. Combust. Inst.* 32 (2009) 847–854.
[3] M.C. Weikl, F. Beyrau, J. Kiefer, T. Seeger, A. Leipertz, *Opt. Lett.* 31 (2006) 1908–1910.
[4] M.C. Weikl, F. Beyrau, A. Leipertz, *Appl. Opt.* 45 (15) (2006) 3646–3651.
[5] P.-E. Bengtsson, L. Martinsson, M. Aldén, S. Kroll, *Combust. Sci. Technol.* 81 (1992) 129–140.
[6] J. Shirley, R. Hall, J. Verdieck, A. Eckbreth, *AIAA Paper* 80-1542 (1979) 1–13.
[7] J. Zheng, J.B. Snow, D.V. Murphy, A. Leipertz, R.K. Chang, R.L. Farrow, *Opt. Lett.* 9 (8) (1984) 341–343.
[8] R.P. Lucht, *Opt. Lett.* 12 (2) (1987) 78–80.
[9] S.P. Kearney, K. Frederickson, T.W. Grasser, *Proc. Combust. Inst.* 32 (2009) 871–878.
[10] M. Aldén, P.-E. Bengtsson, H. Edner, *Appl. Opt.* 25 (23) (1986) 4493–4500.
[11] A.C. Eckbreth, T.J. Anderson, *Opt. Lett.* 11 (8) (1986) 496–498.
[12] T. Seeger, A. Leipertz, *Appl. Opt.* 35 (15) (1996) 2665–2671.
[13] T. Seeger, J. Kiefer, A. Leipertz, B.D. Patterson, C.J. Kliewer, T.B. Settersten, *Opt. Lett.* 34 (2009) 3755.
[14] F. Vestin, M. Afzelius, P.-E. Bengtsson, *Proc. Combust. Inst.* 31 (2007) 833–840.
[15] S. Roy, T.R. Meyer, J.R. Gord, *Appl. Phys. Lett.* 87 (2005) 264103.
[16] B.T. Seeger, J. Kiefer, Y. Gao, B.D. Patterson, C.J. Kliewer, T.B. Settersten, *Opt. Lett.* 35 (2010) 2040–2042.
[17] R.J. Santoro, J.H. Miller, *Langmuir* 3 (2) (1987) 244–254.
[18] H.A. Michelsen, *Appl. Phys. B* 83 (2006) 443–448.
[19] T. Seeger, F. Beyrau, A. Bräuer, A. Leipertz, *J. Raman Spectrosc.* 34 (12) (2003) 932–939.
[20] E. Magens, *Nutzung von Rotations-CARS zur Temperatur-und Konzentrationsmessung in Flammen*, Ph.D. thesis, Universität Erlangen-Nürnberg, 1992.
[21] M.L. Koszykowski, L.A. Rahn, R.E. Palmer, M.E. Coltrin, *J. Chem. Phys.* 91 (1987) 41–46.
[22] L.A. Rahn, R. Palmer, *J. Opt. Soc. Am. B* 3 (1986) 1164–1169.
[23] L. Martinsson, P.-E. Bengtsson, M. Aldén, S. Kröll, J. Bonamy, *J. Chem. Phys.* 99 (4) (1993) 2466–2477.
[24] G. Millot, R. Saint-Loup, J. Santos, R. Chaux, H. Berger, *J. Chem. Phys.* 96 (2) (1992) 961–971.
[25] M. Afzelius, P.-E. Bengtsson, J. Bonamy, *J. Chem. Phys.* 120 (18) (2004) 8616–8623.
[26] M.L. Gonze, R. Saintloup, J. Santos, et al., *Chem. Phys.* 148 (1990) 417.
[27] J. Buldyreva, J. Bonamy, M.C. Weikl, et al., *J. Raman Spectrosc.* 37 (2006) 647–654.
[28] J. Bonamy, D. Robert, J.M. Hartmann, M.L. Gonze, R. Saint-Loup, H. Berger, *J. Chem. Phys* 91 (10) (1989) 5916.
[29] L. Martinsson, P.-E. Bengtsson, M. Aldén, *Appl. Phys. B* 62 (1996) 29–37.
[30] C.S. McEnally, U.O. Koylu, L.D. Pfefferle, D.E. Rosner, *Combust. Flame* 109 (1997) 701–720.